

Lab 4 – Vibe Coding Chat History

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Project: Meteor Dodge

Assigned Repository: SETAPESU26/38-meteor-dodge

Purpose

This document records the Vibe Coding workflow used to inspect, modify, test, and complete the Meteor Dodge project for Lab 4. The work was completed task-by-task and committed separately in the project repository.

Task 1 – Fix Laser Firing / Input State Bug

Problem: SPACE launched the ship from the title screen, but pressing SPACE during active flight did nothing. The required behavior was for SPACE during active flight to fire a defensive laser capable of destroying incoming meteors.

Vibe Coding analysis: The input handling was inspected in `game/game_engine.py`. The game state was represented by `started` and `game_over`. Meteor movement and collision behavior were inspected in `game/meteor.py`.

Implementation: A new `game/Laser.py` entity was added. The engine was updated so SPACE launches from the title screen, fires a laser during active play, and restarts after game over. Lasers move upward, are removed when off-screen, and destroy meteors on collision.

Validation: Syntax checks and runtime checks passed for active firing, laser movement, meteor destruction, off-screen cleanup, and restart behavior.

Commit: Lab 4 Task 1 - Fix laser firing

Task 2 – Implement Meteor Splitting

Problem: Laser–meteor collision originally removed the meteor directly. The required enhancement was for large meteors to split into smaller fragments, while small meteors disappear.

Implementation: The existing meteor radius range was used to distinguish large and small meteors. Large meteors were made to create three smaller child fragments at the parent location. Child velocities combine the parent's movement with outward angular motion so the fragments diverge.

Validation: Syntax and runtime smoke tests passed for large-meteor splitting, small-meteor removal, laser removal after impact, and preservation of earlier behavior.

Commit: Lab 4 Task 2 - Implement meteor splitting

Task 3 – Add Shield Power-up

Problem/feature: A temporary collectible shield power-up was required. The shield should protect the ship from one meteor collision and then be consumed.

Implementation: A new `game/shield.py` entity was created. A shield orb spawns during active gameplay, drifts downward, and can be collected by the ship. Collection activates a temporary shield. A shielded meteor collision removes the meteor, consumes the shield, and allows the run to continue. A visual glow indicates the active shield.

Validation: Syntax and runtime smoke tests passed for shield collection, shield absorption of one meteor hit, and continued gameplay after the protected collision.

Commit: Lab 4 Task 3 - Add shield power-up

Task 4 – Add Survival Multiplier

Feature: A survival-based score multiplier was added. The multiplier begins at x1 and increases after each ten-second survival interval.

Implementation: The engine tracks consecutive survival frames and calculates the multiplier as $1 + \text{consecutive_survival} // (10 * \text{FPS})$. The score accumulates using the current multiplier. An unshielded meteor collision resets the survival streak; a shield-absorbed collision does not reset it.

Validation: Runtime testing verified x1 initially, x2 after 10 seconds, x3 after 20 seconds, reset after an unshielded collision, and preservation of the multiplier after a shield-protected collision.

Commit: Lab 4 Task 4 - Add survival multiplier

Final Verification and Git Workflow

Item	Result
Task 1 – Laser firing	Completed and committed
Task 2 – Meteor splitting	Completed and committed
Task 3 – Shield power-up	Completed and committed
Task 4 – Survival multiplier	Completed and committed
Before video	Recorded as before.mp4
After video	Recorded as after.mp4
Working tree	Clean after final task commit
Final Lab-4 submission	Added to SRN repository

The final Lab-4 submission contains the updated Meteor Dodge source code and the before/after demonstration videos. This document is the required exported chat-history record for the Vibe Coding work.